

Problem Set 4—Suggested Answer

Exercises

1. (a) We evaluate $E(Y | X)$ at each value of X separately, then express the result as a function of X .

- $E(Y | X = 0) = 0$ (mean of $\mathcal{N}(0, 1)$)
- $E(Y | X = 1) = 2$ (mean of $\mathcal{N}(2, 4)$)

Since X takes only the values 0 and 1, a linear function of X that matches both values is

$$E(Y | X) = 2X.$$

Verification: $2 \cdot 0 = 0$ ✓; $2 \cdot 1 = 2$ ✓.

Remark. The conditional expectation $E(Y | X)$ is a *random variable* that is a function of X . Here that function is simply $g(X) = 2X$.

- (b) We evaluate $\text{Var}(Y | X)$ at each value of X :

- $\text{Var}(Y | X = 0) = 1$ (variance of $\mathcal{N}(0, 1)$)
- $\text{Var}(Y | X = 1) = 4$ (variance of $\mathcal{N}(2, 4)$, since the second parameter of $\mathcal{N}(\mu, \sigma^2)$ is the variance)

The linear function $h(X) = a + bX$ satisfying $h(0) = 1$ and $h(1) = 4$ has $a = 1$ and $b = 3$, so

$$\text{Var}(Y | X) = 1 + 3X.$$

Verification: $1 + 3 \cdot 0 = 1$ ✓; $1 + 3 \cdot 1 = 4$ ✓.

Is this homoskedastic? No. Homoskedasticity requires $\text{Var}(Y | X)$ to be constant in X . Here the variance of Y changes with X (it equals 1 when $X = 0$ and 4 when $X = 1$), so the setting is *heteroskedastic*.

- (c) The inner expectation $\text{Var}(Y | X)$ is a random variable that takes the value 1 with probability $\frac{1}{2}$ and the value 4 with probability $\frac{1}{2}$. Using the law of total expectation,

$$E[\text{Var}(Y | X)] = 1 \cdot \Pr(X = 0) + 4 \cdot \Pr(X = 1) = 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{2} = \frac{5}{2}.$$

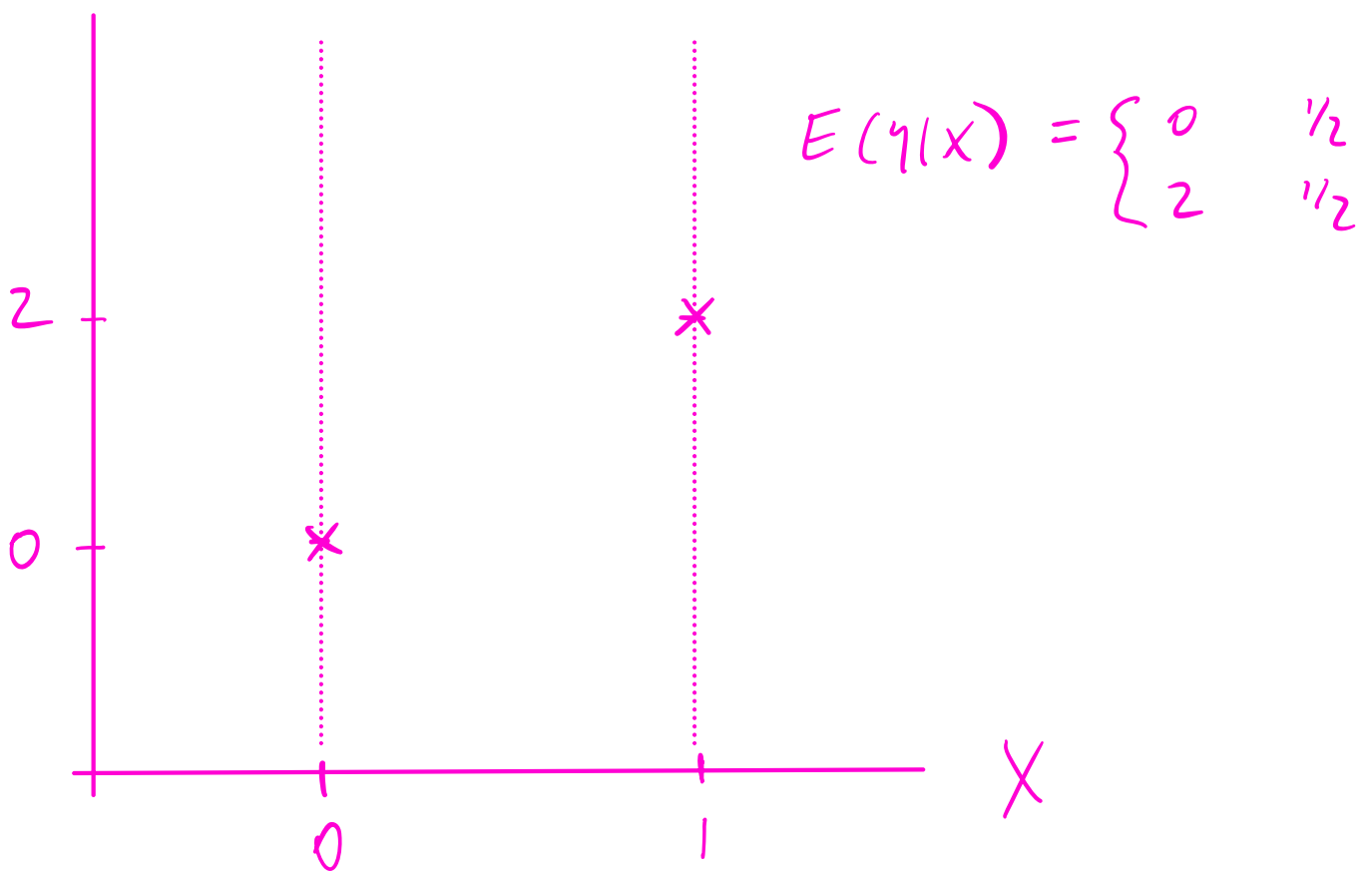
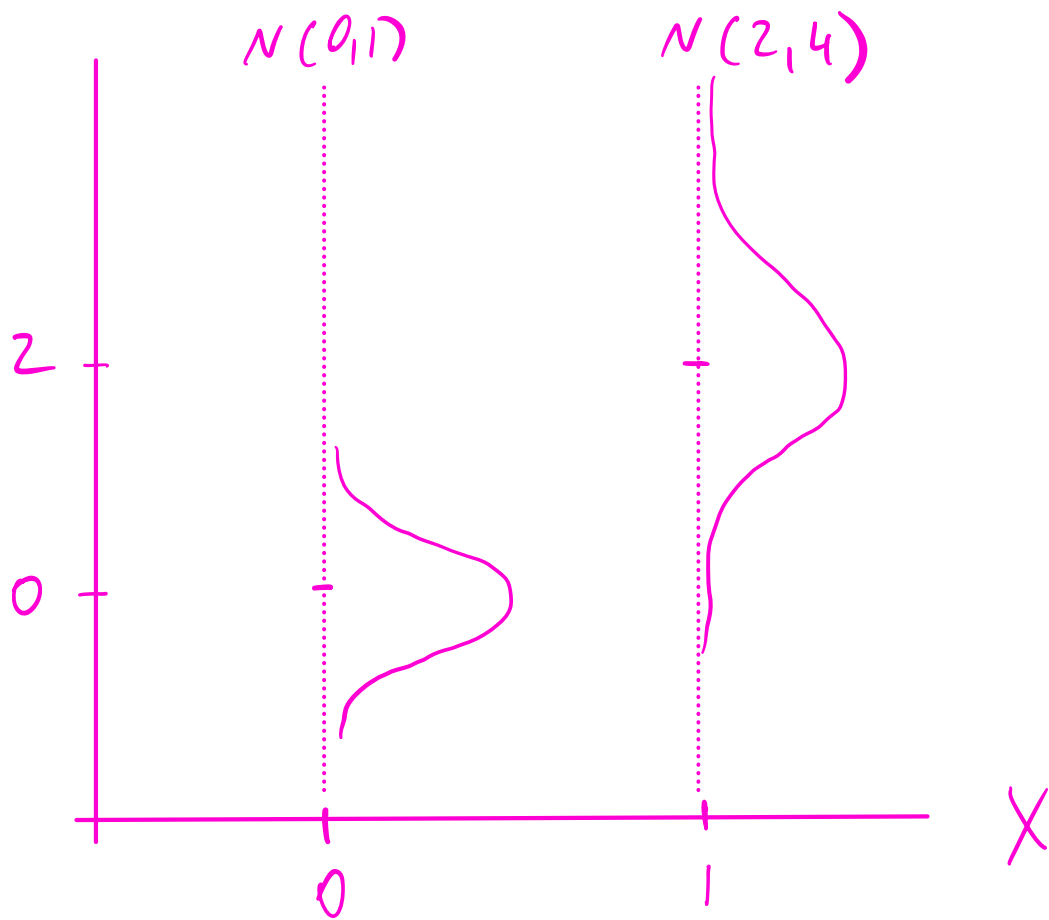
- (d) From part (a), $E(Y | X) = 2X$, so $\text{Var}(E(Y | X)) = \text{Var}(2X) = 4 \text{Var}(X)$.

Since $X \sim \text{Bernoulli}(\frac{1}{2})$, the variance of a $\text{Bernoulli}(p)$ random variable is $p(1-p)$, giving

$$\text{Var}(X) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Therefore

$$\text{Var}(E(Y | X)) = 4 \cdot \frac{1}{4} = 1.$$



(e) Applying the law of total variance directly:

$$\text{Var}(Y) = E[\text{Var}(Y | X)] + \text{Var}(E(Y | X)) = \frac{5}{2} + 1 = \frac{7}{2}.$$

Interpretation. The total variance of Y ($= 3.5$) decomposes into two parts: the *average within-group variance* ($= 2.5$, the average variation of Y within each X -group) and the *between-group variance* ($= 1$, the variation of the conditional mean across X -groups).

2. We want to show that including a constant regressor allows us to assume $E(e_i) = 0$ without loss of generality.

Let $X_{i1} = 1$ for all i (the constant term), and suppose initially that $E(e_i) = c$ for some constant c that may be nonzero. The model is

$$Y_i = X_i' \beta + e_i = \beta_1 + X_{i2} \beta_2 + \cdots + X_{iK} \beta_K + e_i.$$

Add and subtract c :

$$Y_i = \beta_1 + c + X_{i2} \beta_2 + \cdots + X_{iK} \beta_K + (e_i - c).$$



Define $\tilde{\beta}_1 := \beta_1 + c$ and $\tilde{\beta}_j := \beta_j$ for $j = 2, \dots, K$. Then, collecting terms,

$$Y_i = \tilde{\beta}_1 + X_{i2} \tilde{\beta}_2 + \cdots + X_{iK} \tilde{\beta}_K + \tilde{e}_i, \quad \tilde{e}_i := e_i - c.$$

The reparametrized error satisfies

$$E(\tilde{e}_i) = E(e_i - c) = E(e_i) - c = c - c = 0.$$

Because $X_{i1} = 1$, the constant c was absorbed into $\tilde{\beta}_1$. The structural form of the model is unchanged, so we may assume $E(e_i) = 0$ without loss of generality whenever a constant term is included.

Remark. This is the key reason we always include a constant in regression. If we omitted it, $E(e_i)$ could be nonzero, biasing the slope estimates.

3. **Claim.** $\hat{\beta}^{\text{OLS}} \xrightarrow{P} \beta$ as $N \rightarrow \infty$.

Proof. Starting from $Y = X\beta + e$ and the OLS formula,

$$\hat{\beta}^{\text{OLS}} = (X'X)^{-1}X'Y = (X'X)^{-1}X'(X\beta + e) = \beta + (X'X)^{-1}X'e.$$

Multiply and divide by N :

$$\hat{\beta}^{\text{OLS}} - \beta = \left(\frac{X'X}{N} \right)^{-1} \frac{X'e}{N}.$$

$$\begin{aligned} E(e_i | X_i) &= E(e_i) = 0 \\ \Rightarrow E(e_i X_i) &= 0 \\ &\text{by LIE} \end{aligned}$$

We analyze each factor separately.

First factor: $\left(\frac{X'X}{N} \right)^{-1} = \mathbf{O}_p(1).$

By the Law of Large Numbers,

$$\frac{X'X}{N} = \frac{1}{N} \sum_{i=1}^N X_i X_i' \xrightarrow{p} M_{XX} := E(X_i X_i'),$$

where M_{XX} is finite and (by assumption) has full rank. Therefore $(X'X/N)^{-1} \xrightarrow{p} M_{XX}^{-1}$, which is a fixed finite matrix, hence $(\frac{X'X}{N})^{-1} = O_p(1)$.

Second factor: $\frac{X'e}{N} = o_p(1)$.

The exogeneity assumption gives $E(X_i e_i) = 0$. By the Law of Large Numbers,

$$\frac{X'e}{N} = \frac{1}{N} \sum_{i=1}^N X_i e_i \xrightarrow{p} E(X_i e_i) = 0,$$

which means $\frac{X'e}{N} = o_p(1)$.

Conclusion. Using the product rule for stochastic orders,

$$\hat{\beta}^{\text{OLS}} - \beta = \underbrace{\left(\frac{X'X}{N}\right)^{-1}}_{O_p(1)} \cdot \underbrace{\frac{X'e}{N}}_{o_p(1)} = o_p(1).$$

Since $\hat{\beta}^{\text{OLS}} - \beta = o_p(1)$, we have $\hat{\beta}^{\text{OLS}} \xrightarrow{p} \beta$. □

4. The model is $Y = X\beta + e$ with $E(e \mid X) = 0$ and $E(ee' \mid X) = \sigma^2 \Gamma = \Sigma$, where $\Gamma = \text{diag}(\gamma_1, \dots, \gamma_N)$ and $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_N^2)$ with $\sigma_i^2 = \sigma^2 \gamma_i$.

(a) From the OLS formula, $\hat{\beta}^{\text{OLS}} - \beta = (X'X)^{-1} X'e$. Therefore,

$$\begin{aligned} \text{Var}(\hat{\beta}^{\text{OLS}} \mid X) &= (X'X)^{-1} X' \text{Var}(e \mid X) X (X'X)^{-1} \\ &= (X'X)^{-1} X' \Sigma X (X'X)^{-1}. \end{aligned}$$

This is often called the *sandwich* (or *Eicker–Huber–White*) variance. It differs from the classical OLS formula $\sigma^2(X'X)^{-1}$ because the error variances σ_i^2 are not all equal.

(b) We prove that every linear unbiased estimator has variance at least $(X'\Sigma^{-1}X)^{-1}$ in the matrix (positive semidefinite) sense.

Let $\tilde{\beta} = C'Y$ where C is $N \times K$ and may depend on X .

Step 1: Unbiasedness condition.

$$E(\tilde{\beta} \mid X) = C'E(Y \mid X) = C'X\beta = \beta \quad \forall \beta \implies C'X = I_K.$$

Step 2: Variance of $\tilde{\beta}$.

$$\text{Var}(\tilde{\beta} \mid X) = C' \text{Var}(Y \mid X) C = C' \Sigma C.$$

Step 3: Decompose C' . Define

$$D' := C' - (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1},$$

so that $C' = D' + (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}$.

Note that by the unbiasedness condition $C'X = I_K$,

$$D'X = C'X - (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}X = I_K - I_K = 0.$$

Step 4: Expand $\text{Var}(\tilde{\beta} | X) = C'\Sigma C$.

Substituting $C' = D' + (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}$ and $C = D + \Sigma^{-1}X(X'\Sigma^{-1}X)^{-1}$,

$$\begin{aligned} C'\Sigma C &= (D' + (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1})\Sigma(D + \Sigma^{-1}X(X'\Sigma^{-1}X)^{-1}) \\ &= D'\Sigma D + D'\Sigma\Sigma^{-1}X(X'\Sigma^{-1}X)^{-1} + (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\Sigma D \\ &\quad + (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\Sigma\Sigma^{-1}X(X'\Sigma^{-1}X)^{-1} \\ &= D'\Sigma D + \underbrace{D'X(X'\Sigma^{-1}X)^{-1}}_{=0} + (X'\Sigma^{-1}X)^{-1}\underbrace{X'D}_{=0} + (X'\Sigma^{-1}X)^{-1} \\ &= D'\Sigma D + (X'\Sigma^{-1}X)^{-1}. \quad \geq (X'\Sigma^{-1}X)^{-1} \end{aligned}$$

Step 5: Conclude. Since Σ is positive (semi)definite, $D'\Sigma D \geq 0$ in the positive semidefinite sense. Therefore

$$\text{Var}(\tilde{\beta} | X) = D'\Sigma D + (X'\Sigma^{-1}X)^{-1} \geq (X'\Sigma^{-1}X)^{-1}.$$

The lower bound $(X'\Sigma^{-1}X)^{-1}$ is achieved if and only if $D = 0$, i.e., $C' = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}$.
□

(c) **Deriving $\text{Var}(\hat{\beta}_{GLS} | X)$.**

By definition,

$$\hat{\beta}_{GLS} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\tilde{Y} = (X'\Gamma^{-1}X)^{-1}X'\Gamma^{-1}Y.$$

Substituting $Y = X\beta + e$,

$$\hat{\beta}_{GLS} - \beta = (X'\Gamma^{-1}X)^{-1}X'\Gamma^{-1}e.$$

Therefore,

$$\begin{aligned} \text{Var}(\hat{\beta}_{GLS} | X) &= (X'\Gamma^{-1}X)^{-1}X'\Gamma^{-1} \text{Var}(e | X) \Gamma^{-1}X(X'\Gamma^{-1}X)^{-1} \\ &= (X'\Gamma^{-1}X)^{-1}X'\Gamma^{-1}(\sigma^2\Gamma)\Gamma^{-1}X(X'\Gamma^{-1}X)^{-1} \\ &= \sigma^2(X'\Gamma^{-1}X)^{-1}X'\Gamma^{-1}X(X'\Gamma^{-1}X)^{-1} \\ &= \sigma^2(X'\Gamma^{-1}X)^{-1}. \end{aligned}$$

Since $\Sigma = \sigma^2\Gamma$ implies $\Sigma^{-1} = \frac{1}{\sigma^2}\Gamma^{-1}$, we have $X'\Sigma^{-1}X = \frac{1}{\sigma^2}X'\Gamma^{-1}X$, and hence

$$(X'\Sigma^{-1}X)^{-1} = \sigma^2(X'\Gamma^{-1}X)^{-1}.$$

Therefore

$$\boxed{\text{Var}(\hat{\beta}_{GLS} | X) = \sigma^2(X'\Gamma^{-1}X)^{-1} = (X'\Sigma^{-1}X)^{-1}.}$$

Comparison with part (b). The GLS estimator achieves exactly the lower bound from part (b). In other words, among all linear unbiased estimators, GLS has the *smallest* (matrix) variance. This is the *Gauss–Markov theorem for the heteroskedastic case*: GLS is BLUE.

Intuition. GLS downweights observations with large variance (γ_i large) and upweights those with small variance (γ_i small), extracting the maximum information from the data. OLS treats all observations equally and therefore wastes information in a heteroskedastic setting.